Budgets, Banking and Simple Interest

Concept

A budget matches income, expenses and savings over the same time period.

Balance

Income − expenses = surplus or deficit.

Transactions

Add deposits; subtract withdrawals and fees.

Simple interest

$I = Prt$; $A = P + I$. Annual r needs t in years.

Savings goal

(Goal − current savings)/number of periods.

Example 1

Find simple interest on $900 at 4% annually for 8 months.

Solution

1. $t = 8/12$ year; $r = 0.04$.

2. $I = 900(0.04)(8/12) = \$24$; $A = \$924$.

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Example 2

You need $480 in 6 months and have $150. How much must you save monthly?

Solution

1. Amount still needed = 480 − 150 = $330.

2. Monthly saving = 330/6 = $55, ignoring interest.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the surplus if income is $850 and expenses total $735.

2. Find simple interest on $1200 at 3.5% annually for 9 months.

3. Convert a $15 weekly expense to an average monthly expense.

4. A $720 goal is 8 months away and $200 is saved. Find monthly saving needed.

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Practice answers | check after trying

1. $115.

2. 1200(0.035)(9/12) = $31.50; final amount $1231.50.

3. 15 × 52/12 = $65 per average month.

4. (720 − 200)/8 = $65 per month. If the budget cannot support this, adjust the timeline, goal or other expenses.